

CAYO COCO, Cuba

WMO: 782390

Lat: 22.539N

Lon: 78.368W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (CUB)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 60.3	(c) 63.4	(d) 49.0	(e) 51.4	(f) 65.6	(g) 52.0	(h) 57.5	(i) 68.1	(j) 25.7	(k) 72.1	(l) 23.4	(m) 72.3	(n) 4.2	(o) 10	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	90.8	78.0	90.0	78.0	89.2	77.9	80.0	87.6	79.6	87.1	79.1	86.5	11.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.0	145.7	84.1	77.4	142.5	83.8	76.9	140.5	83.6	43.5	88.1	43.0	87.3	42.5	86.7	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.7	(b) 20.5	(c) 18.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 22.7	(5) 20.5	(6) 18.6	(7) 51.3	(8) 80.8	(9) 2.4	(10) 1.0	(11) 49.6	(12) 81.5	(13) 48.2	(14) 82.1	(15) 46.9	(16) 82.6	(17) 45.1	(18) 83.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.0	73.5	74.8	75.5	78.6	79.8	82.3	83.6	83.7	82.5	80.5	77.1	75.5
	DBStd	4.47	3.85	3.42	3.28	2.44	2.43	2.28	1.86	1.82	1.74	2.29	3.06	3.97
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	5	2	1	0	0	0	0	0	0	0	0	0	2
	CDD50	10581	730	694	791	859	924	970	1041	1045	976	947	813	791
	CDD65	5112	267	275	327	409	459	520	576	580	526	482	363	328
	CDH74	47177	1212	1506	1890	3297	4309	5854	6983	7014	5913	4683	2534	1982
	CDH80	14165	103	178	297	684	1113	2118	2855	2897	2147	1235	344	194
Wind	WSAvg	9.7	10.3	10.6	11.1	10.7	9.3	7.9	8.9	8.9	7.8	8.9	11.0	10.5
Precipitation	PrecAvg	36.7	1.2	1.2	1.2	1.9	4.0	5.4	4.0	4.8	5.4	4.3	2.0	1.4
	PrecMax	53.5	2.4	3.3	3.9	5.0	11.8	12.9	7.4	6.8	7.6	9.3	6.5	4.6
	PrecMin	22.5	0.2	0.2	0.2	0.4	0.6	1.5	0.9	2.5	3.1	1.4	0.5	0.1
	PrecStd	6.6	0.6	0.8	0.8	1.0	2.6	2.5	1.4	1.1	1.2	2.1	1.4	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.3	85.7	88.1	88.8	89.5	91.9	91.8	91.8	90.9	89.1	86.3	84.9
		MCWB	72.3	71.3	72.9	73.7	76.5	77.7	78.2	78.1	78.4	77.3	77.4	75.0
	2%	DB	82.5	83.6	85.1	86.2	87.6	90.0	90.7	90.6	89.9	87.8	84.8	83.2
		MCWB	73.0	72.2	73.1	74.1	76.4	77.7	78.1	78.4	78.5	77.6	76.3	74.7
	5%	DB	81.0	82.2	83.2	85.1	86.2	88.9	89.7	89.8	88.8	86.7	83.3	82.1
		MCWB	72.5	72.5	72.5	73.9	75.8	77.7	77.9	78.4	78.2	77.4	75.3	74.4
	10%	DB	79.6	80.7	81.5	84.0	85.1	87.7	88.8	88.9	87.8	85.6	81.9	81.0
		MCWB	71.8	71.8	71.5	73.6	75.2	77.7	77.7	78.2	78.0	77.1	74.4	73.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.5	75.5	76.1	77.4	79.0	80.4	80.2	80.9	80.8	79.9	78.3	77.8
		MCDB	80.6	80.6	81.8	83.0	86.3	88.0	88.8	88.7	88.1	86.2	84.2	82.2
	2%	WB	74.5	74.4	74.9	76.3	77.9	79.5	79.6	80.0	79.9	79.1	77.5	76.4
		MCDB	79.8	80.0	81.3	83.0	84.5	86.9	87.9	87.8	87.1	85.4	82.9	80.7
	5%	WB	73.6	73.6	73.9	75.4	77.2	78.8	79.0	79.3	79.2	78.3	76.6	75.5
		MCDB	79.2	79.7	80.8	82.4	83.3	86.2	87.1	87.0	86.3	84.5	81.7	80.3
	10%	WB	72.8	72.8	72.8	74.6	76.5	78.2	78.4	78.9	78.6	77.7	75.8	74.6
		MCDB	78.3	79.0	79.9	81.5	82.6	85.4	86.3	86.5	85.6	83.8	80.8	79.6
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.8	10.9	10.4	10.3	10.9	10.8	10.8	10.9	9.3	8.8	9.4
		MCDBR	11.9	13.2	13.7	12.1	11.8	12.4	12.1	11.5	11.9	11.0	9.6	9.9
		MCWBR	5.4	5.7	5.2	4.6	4.4	4.3	4.0	4.0	4.1	4.1	4.2	4.7
	5% WB	MCDBR	11.2	10.8	11.8	10.7	10.2	11.5	11.1	11.2	11.1	10.3	8.4	9.4
MCWBR		5.5	5.4	5.0	4.7	4.3	4.6	4.3	4.4	4.3	4.3	4.0	4.7	
Clear-Sky Solar Irradiance	taub	0.364	0.373	0.388	0.409	0.434	0.469	0.474	0.450	0.421	0.413	0.389	0.386	
	taud	2.571	2.541	2.489	2.421	2.397	2.324	2.297	2.393	2.489	2.499	2.520	2.519	
	Ebn at Noon	287	291	290	284	274	263	261	269	276	274	276	275	
	Edh at Noon	29	32	35	38	38	41	42	38	34	33	30	30	
All-Sky Solar Radiation	RadAvg	1271	1570	1817	2040	1981	1894	2052	1949	1748	1497	1318	1205	
	RadStd	80	65	73	72	163	117	115	78	78	97	76	85	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page